Big Data and Machine Learning for Applied Economics

Week 06: Opening the Black Box: What the Machine Learned

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Previously...

- **Split, average, and correct:** a tree splits the stations into groups and averages; a forest averages hundreds of trees; boosting adds small trees, each correcting the previous one.
- **Week 5's answer:** with the best line's same ingredients (census and department), boosting left 12.8 against 15.6. With single-split trees, no interactions, the judge gave 15.5; with six, 13.2: the interactions the machine found are worth 2.3 points.
- **The two leagues, together:** averaging the boosting and the map's k-NN left 11.4, the course's best.

What was left pending

Boosting predicts better than everything we wrote by hand, but it does not say what it learned: which variables it uses, with what shape, which interactions it found, and why it predicts what it predicts at each station.

Today's question: what did the machine learn?

No new model enters today: we enter week 5's

- **The model:** 738 trees with 38,330 leaves in total. Nobody reads it, and it predicts a station's vote with 12.8 points of error.
- **Four questions, four tools:** which variables does it use? (importance); with what shape? (partial dependence); why does it predict what it predicts at a station? (Shapley and SHAP); who does it fail? (the error by subgroup).
- **And a fifth, the most important:** what do these tools *not* tell us?

The takeaway

Each of today's tools moves one variable and watches what the prediction does. It asks the model, not the world: it says how it predicts, not what would happen if the world changed.

1. Which variables does it use? Importance
2. With what shape? Partial dependence
3. Why this station? Shapley and SHAP
4. When the map comes in: the explanation belongs to the model
5. Who does it fail?
6. Problem Set 2
7. Closing

The model we are about to open

Week 5's boosting: the 37 census variables and the department, in seven blocks

Block	Columns	What it says about the station's surroundings
Education	3	no education, technical, and higher (% of people)
Age and gender	7	five age groups, women, and voting-age
Ethnicity	5	Indigenous, Rom, Raizal, Palenquero, and Afro
Utilities	5	water, sewerage, power, gas, and internet (% of housing units)
Stratum	7	housing units in each stratum, 0 through 6
Scale and rurality	10	population, housing, blocks, households, distance, registered voters, zone
Department	33	one column of zeros and ones per department

- **Three quantities, all session long:** $\hat{f}(x)$, a station's prediction; $\mathbb{E}[\hat{f}(X)]$, the average prediction or *base* (52.5); and the error **on the test set**: interpreting a model on the data it memorized is interpreting its memory.
- **Notation:** x_{-j} is every variable except j. Nearly everything today is "move x_j and watch what $\hat{f}$ does".

The importance the model ships with: gain

How much the training error fell at the splits that used each variable, across the 738 trees

- **Definition:** every split lowers the training loss; x_j's gain adds those drops over all splits on x_j (`xgb.importance`). It is the forest's impurity importance.
- **In our model:** the department (its 33 columns added up) takes 29.1 % of the gain; the Afro-descendant share, 14.5 %; the Indigenous share, 11.8 %.

Question for the room

If we add a column of random numbers to the model, how much gain does it receive?

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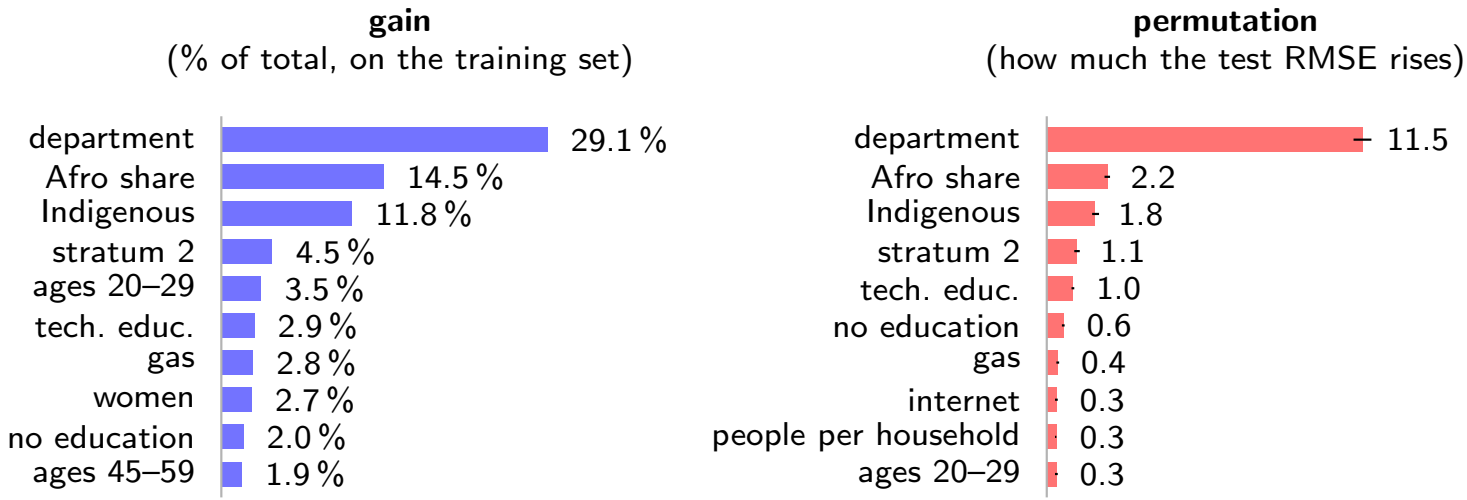
If we add a column of random numbers to the model, how much gain does it receive?

We did it: retrained with a noise column, the noise lands at rank 24 of 39 (0.9 % of the gain), above 15 real variables (rurality lands at rank 39). A continuous variable offers thousands of thresholds, and in some small node the noise "helps" separate the training data: the cardinality bias (Strobl et al., 2007). By permutation, on the test set, the same noise is worth 0.01 points.

Permutation importance: how much shuffling hurts

Breiman (2001); Fisher, Rudin, and Dominici (2019). On the test set, ten shuffles per variable

$$\mathrm{VI}_j \;=\; \mathrm{RMSE}\big(y, \hat{f}(X^{(j)})\big) \;-\; \mathrm{RMSE}\big(y, \hat{f}(X)\big)$$



- $X^{(j)}$ is the test set with x_j shuffled: its link with y is broken, with no retraining; if the error does not rise, the model did not need it. Black lines: one standard deviation across the ten shuffles. The department is shuffled whole, its 33 columns at once.
- Shuffling the department raises the error 11.5 points; the Afro share, 2.2; the Indigenous share, 1.8. Higher education, week 2's star, barely 0.25. The two importances agree at the top and part ways below.

Importance's traps

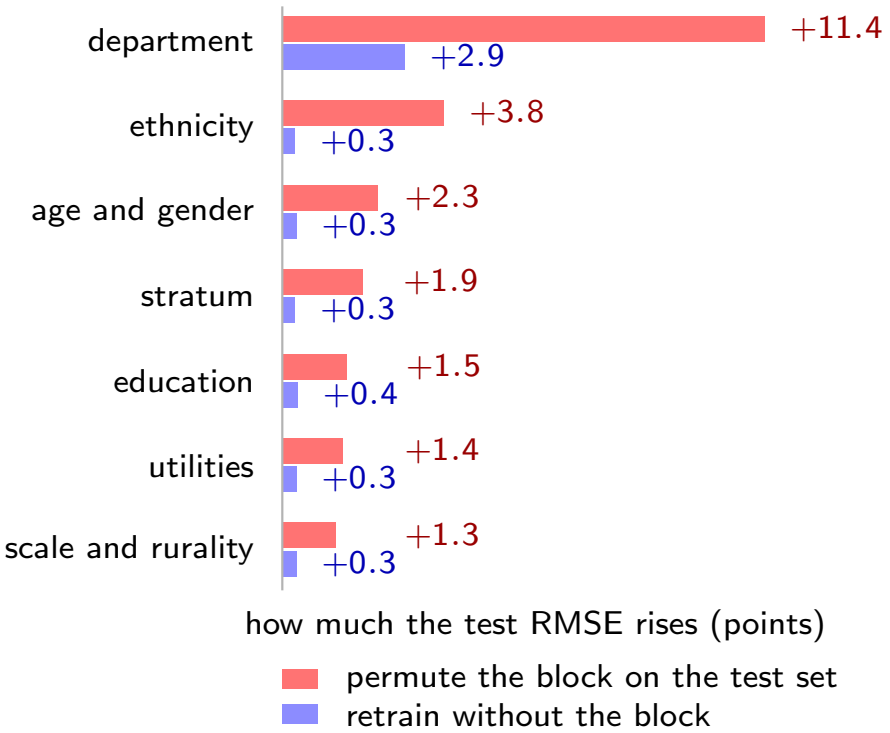
1. **Permuting forces extrapolation** (Hooker, Mentch, and Zhou, 2021). Shuffling the Afro share while holding the department fixed manufactures Chocó stations with no Afro population, which do not exist.
2. **Credit gets split.** Two variables telling the same story look small on their own and large together: they take turns at the splits and each takes half.
3. **It depends on the model.** Two models with the same error can lean on different variables (part 4).
4. **It is unstable.** It changes with the seed, the resample, and the shuffles: report it with bars.

And the usual one

Important is not causal. The most important variable may have no effect at all: it only needs to be the best proxy for what does.

Permuting versus retraining: the blocks

Permutation asks about the model we have; ablation, about the best model without that block



- **Permuting a census block** raises the error between 1.3 and 3.8 points: the model uses them all.
- **Retraining without it** costs only 0.3 to 0.4: the other blocks tell the same story and stand in for it.
- **The department** is not fully replaced: permuting it costs 11.4 and removing it, 2.9.
- **The information comes bundled.** How many distinct things do the 37 columns say? Week 7 compresses them.

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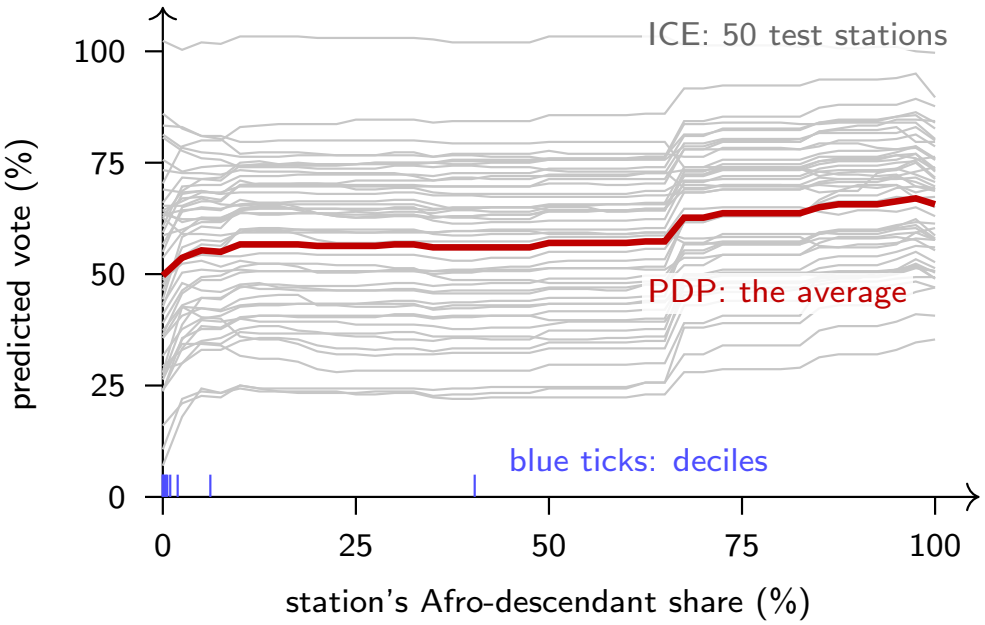
6 Problem Set 2

7 Closing

Partial dependence and its individual curves

Friedman (2001); Goldstein et al. (2015). The Afro-descendant share, fixed at each value for the 2,398 test stations

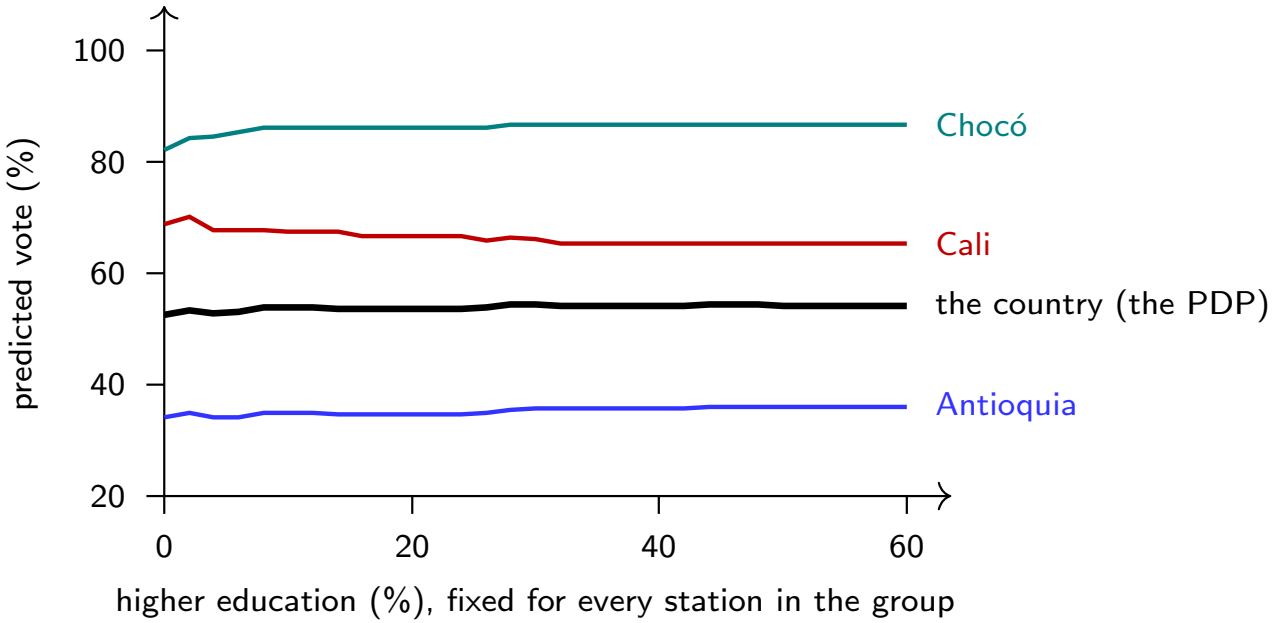
$$\text{PDP:} \quad \hat{f}_j(z) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}\left(z, x_{i,-j}\right) \qquad \text{ICE}_i(z) = \hat{f}\left(z, x_{i,-j}\right)$$



- **PDP:** fix $x_j = z$ at every station, predict, and average: from 49.6 at 0 % Afro share to 65.8 at 100 %.
- **ICE:** the same, one curve per station. Nearly parallel: the Afro share adds the same on top of each station's level.
- **Blue ticks:** the deciles; 80 % of the stations have less than 6.2 % Afro share.

Cali's star, country-wide

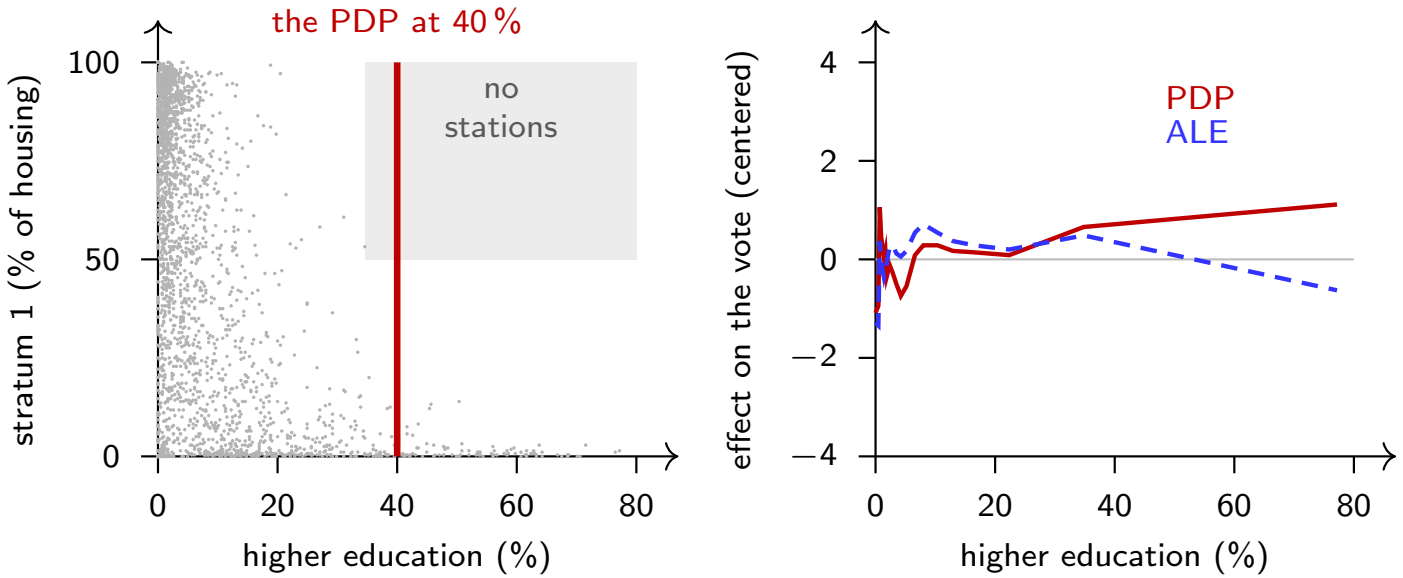
Higher education: the country's curve and the average ICE curves of three groups of stations



- **Country-wide, nearly flat:** the PDP runs from 52.4 to 54.4. Week 2's star variable barely moves the prediction.
- **But the average hides opposite slopes:** at Cali's 43 test stations, raising education *lowers* the prediction (from 68.8 to 65.2); at Chocó's 79, it *raises* it (from 82.1 to 86.6). Slopes that flip sign with place are an interaction: week 3's funnel, found without writing it.

When the PDP asks about stations that do not exist

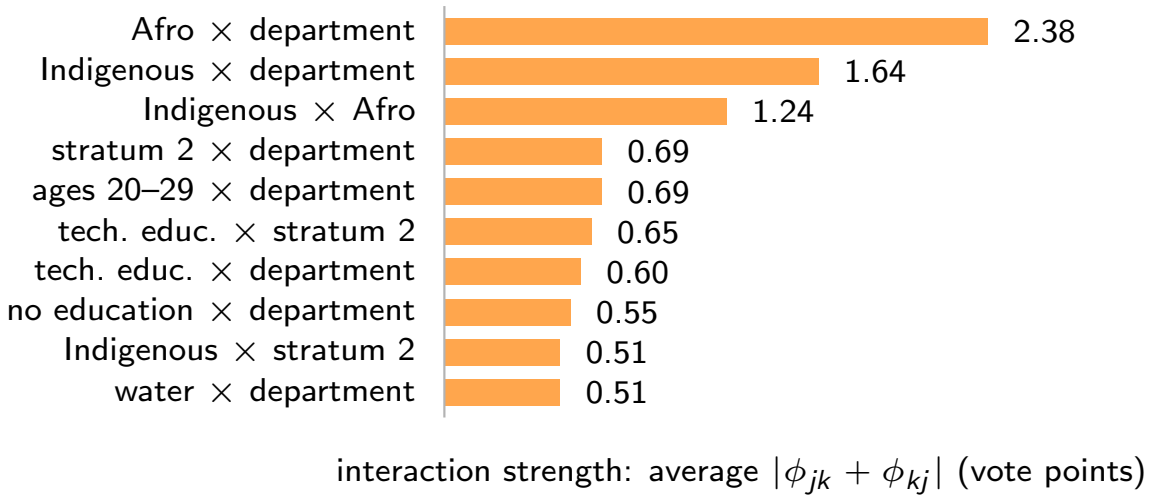
Left: higher education against stratum 1. Right: the education PDP and ALE (Apley and Zhu, 2020)



- **The PDP at 40 %** sets all 2,398 stations at 40 % higher education, including the 1,208 (50 %) with more than half their housing in stratum 1. No such station exceeds 34.7 %: the model predicts for stations that do not exist. It is week 2's extrapolation, hidden inside an average.
- **The ALE** looks only at local differences: within each education band, how much $\hat{f}$ changes between the edges for the stations actually there, then accumulates (appendix A4). This time they agree (maximum difference: 1.7 points); with more correlated variables, the PDP can invent a shape.

The interactions the machine found

SHAP interaction values at 500 test stations: the credit that only exists when two variables move together



- A station's SHAP split divides into each variable's own effect and one per pair, ϕ_{jk}: what the model does with x_j *because* it saw x_k (Lundberg et al., 2020; appendix A6).
- **The answer to week 5's pending question:** the interactions worth 2.3 points are, above all, ethnicity within each department: exactly the slopes week 3 refused to write by hand and the machine found here on its own.

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Is it Chocó or is it the Afro-descendant share?

A station in Quibdó, three players, and one regression per coalition

Colegio Pedro Grau Arola (Quibdó; voted 88.8 for Petro). The game: $v(S)$ is the prediction of a regression with the blocks in S, minus the average (52.5). The players: **E**ducation, e**T**hnicity, and the **D**epartment.

S	$\emptyset$	E	T	D	ET	ED	TD	ETD
$v(S)$	0.0	-1.1	31.2	33.0	33.2	34.7	32.2	34.6

Order of entry	E adds	T adds	D adds
E, T, D	-1.1	34.3	1.4
E, D, T	-1.1	-0.1	35.8
T, E, D	2.1	31.2	1.4
T, D, E	2.4	31.2	1.1
D, E, T	1.7	-0.1	33.0
D, T, E	2.4	-0.8	33.0
Average, ϕ	**1.1**	**15.9**	**17.6**

Shapley's idea (1953): let the players enter in some order, write down how much each adds on entry, and average over all orders.

Question for the room: how much of this station's vote is Chocó and how much is the Afro-descendant share?

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It depends on the order: whoever enters first takes almost everything (31.2 or 33.0 points) and the second, almost nothing, because they tell the same story. Shapley splits it: 15.9 to ethnicity, 17.6 to the department, and 1.1 to education; they add to 34.6, the distance between the prediction (87.1) and the average.

The Shapley value: the formula and why it is unique

$$\phi_j \;=\; \sum_{S \subseteq \{1,\dots,p\}\setminus\{j\}} \frac{|S|!\,(p-|S|-1)!}{p!}\,\Big[v(S\cup\{j\})-v(S)\Big]$$

The weight is the fraction of orders in which j enters right after those in S: the previous table, written in general.

It is the only splitting rule with four reasonable properties (appendix A1):

- **Efficiency:** $\sum_j \phi_j = v(\text{all}) - v(\emptyset)$. All the credit is handed out.
- **Symmetry:** two interchangeable players receive the same.
- **Null player:** whoever never adds anything receives zero.
- **Additivity:** the split of the sum of two games is the sum of the splits.

From blocks to columns

With p columns there are 2^p coalitions: with our model's 70, a 22-digit number. And you must decide what "using only those in S" means without retraining 2^p models.

SHAP: Shapley for machine learning models

Lundberg and Lee (2017); Lundberg et al. (2020)

The local explanation is additive: for a station x,

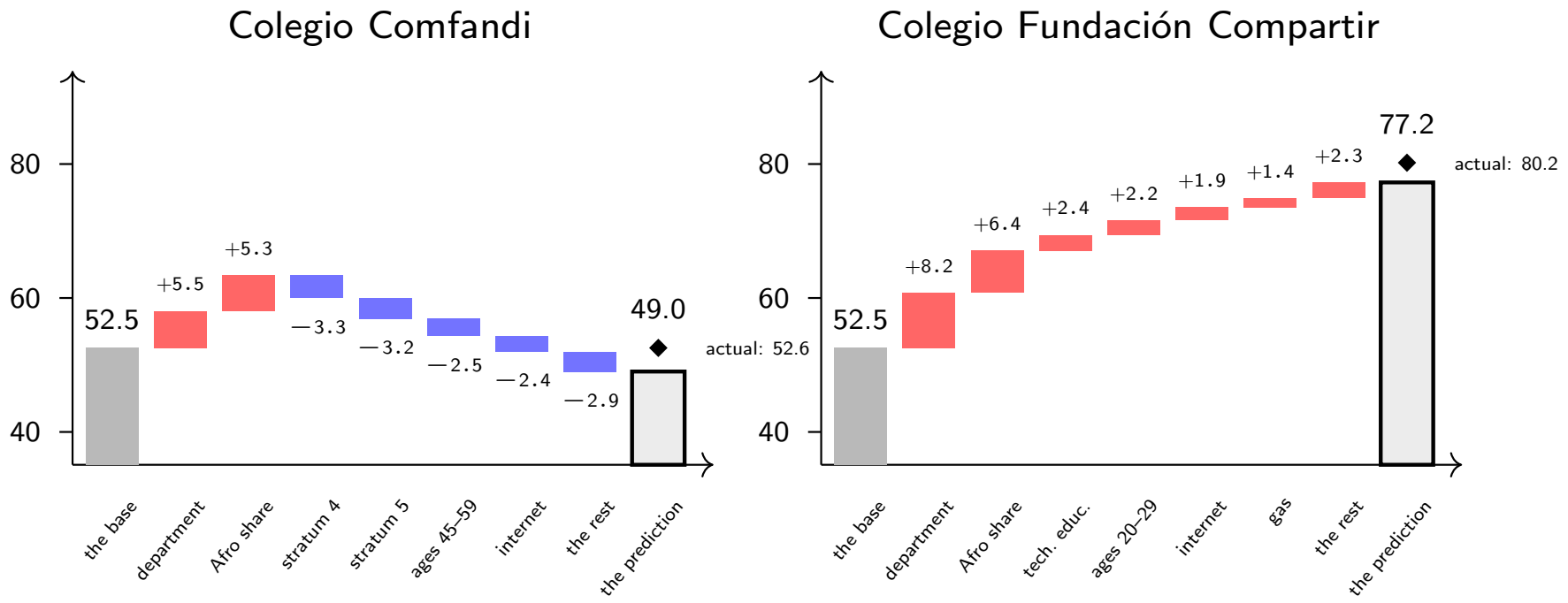
$$\hat{f}(x) = \underbrace{\mathbb{E}[\hat{f}(X)]}_{\phi_0,\ \text{the base}} + \sum_{j=1}^{p} \phi_j(x),$$

and each $\phi_j(x)$ is column j's Shapley value in the game "predict x".

- **What $v(S)$ is for a model:** the expected prediction when only the columns in S are known; the absent ones are filled with background data, ignoring the correlation (*marginal*) or honoring it (*conditional*). Appendix A2.
- **How it is computed:** KernelSHAP samples coalitions (any model; slow; appendix A3). **TreeSHAP** is exact and fast on trees: the 2,398 test stations in seconds (`predcontrib = TRUE` in `xgboost`).
- **Verified station by station:** base (52.5) + sum of contributions = prediction, with a maximum error of 0.0002 points (machine rounding).

Two schools in Cali, 28 points apart

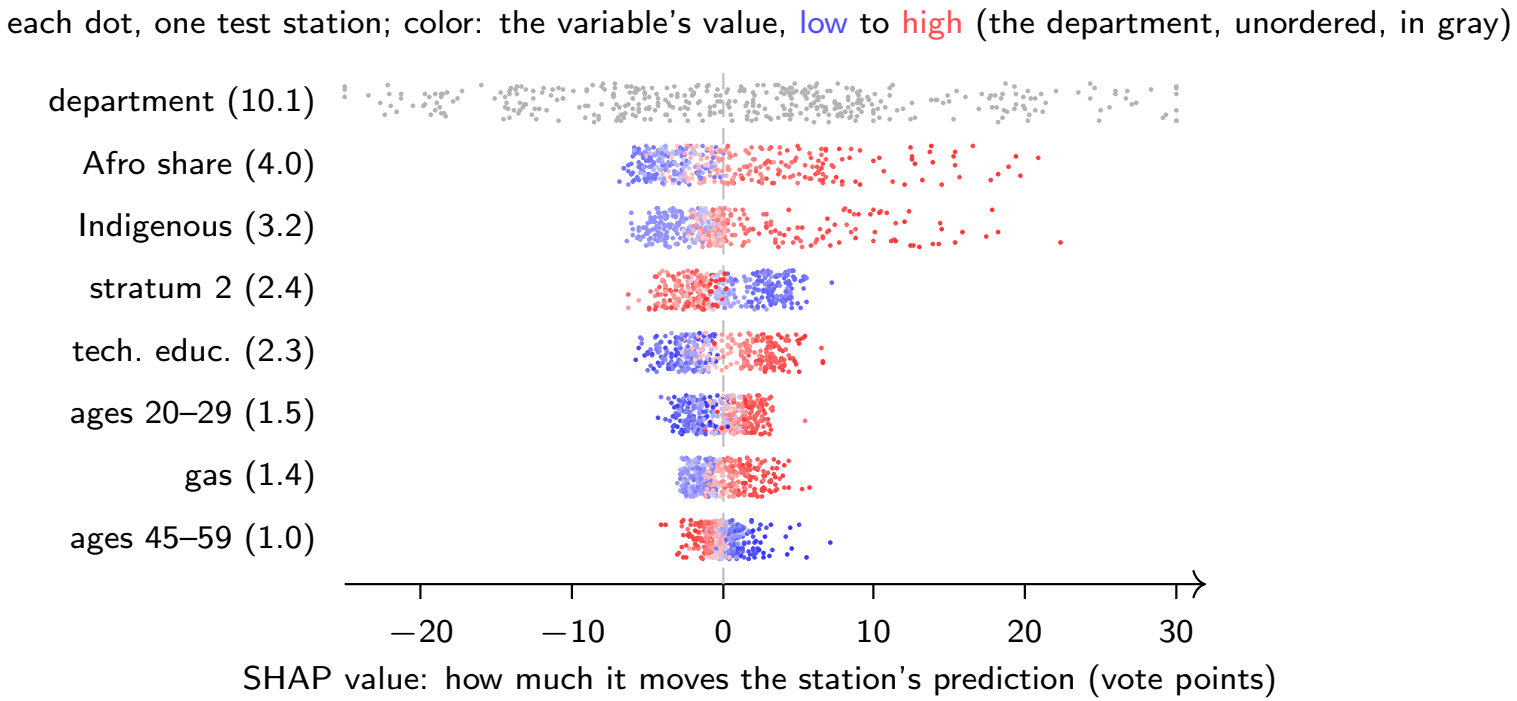
Each prediction's SHAP split, from the base to the prediction: the six columns that move it most



- **Colegio Comfandi:** 41 % higher education and 78 % of housing in strata 4 and 5, which pull it down. **Colegio Fundación Compartir:** 71 % in stratum 1 and a 35 % Afro share.
- **The same department weighs differently** (5.5 and 8.2): with no interactions it would weigh the same in both. It is an interaction, seen at one station.

From local to global: every station at once

The beeswarm: one row per variable, one dot per test station; in parentheses, the average |SHAP|



- **Global importance** is the average absolute value: department 10.1, Afro share 4.0, Indigenous 3.2. The same hierarchy as permutation, with a local reading behind it.
- **Color gives the direction** and **the spread within a row is the interaction:** the same variable value pushes differently depending on the rest of the station.

What a SHAP value is not

1. **It is not a causal effect.** It splits a prediction across the model's columns: the Afro share does not "cause" its points; it is the column that best summarizes what the model learned about the Pacific.
2. **It is not unique.** It changes with the fill-in, the background, and the model (part 4): report what it was computed with.
3. **It is not about the voter.** The station is an aggregate: ecological inference (Robinson, 1950).
4. **It is not a forecast.** It explains the 2022 election; that it repeats in 2026 is a hypothesis.
5. **It does not replace the test set.** A pretty explanation of a bad model is a story about noise.

Rudin (2019), in one sentence

Explaining a black box *afterwards* does not make it transparent: when the decision demands it, compare with a model interpretable by design and measure how much is lost.

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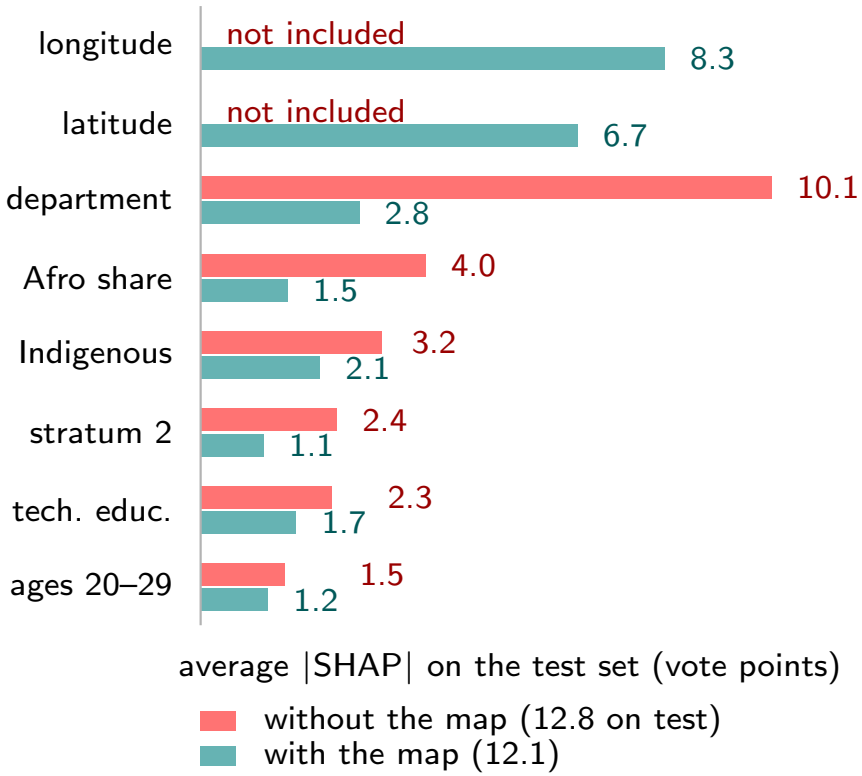
5 Who does it fail?

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The same problem, another explanation

Week 5's boosting with coordinates: almost the same error, another split of the credit



- **With the coordinates,** the error falls from 12.8 to 12.1 and the credit moves out: they take what belonged to the department (from 10.1 to 2.8) and to the Afro share (from 4.0 to 1.5).
- **Which one is the true one?** Neither: two nearly equally good models, two stories of the same vote: the Rashomon effect (Breiman, 2001; Fisher, Rudin, and Dominici, 2019).
- **The census, given the map:** without it, 12.6; with it, 12.1.
- **The rule:** an explanation belongs to one model; ask whether another just as good would say the same.

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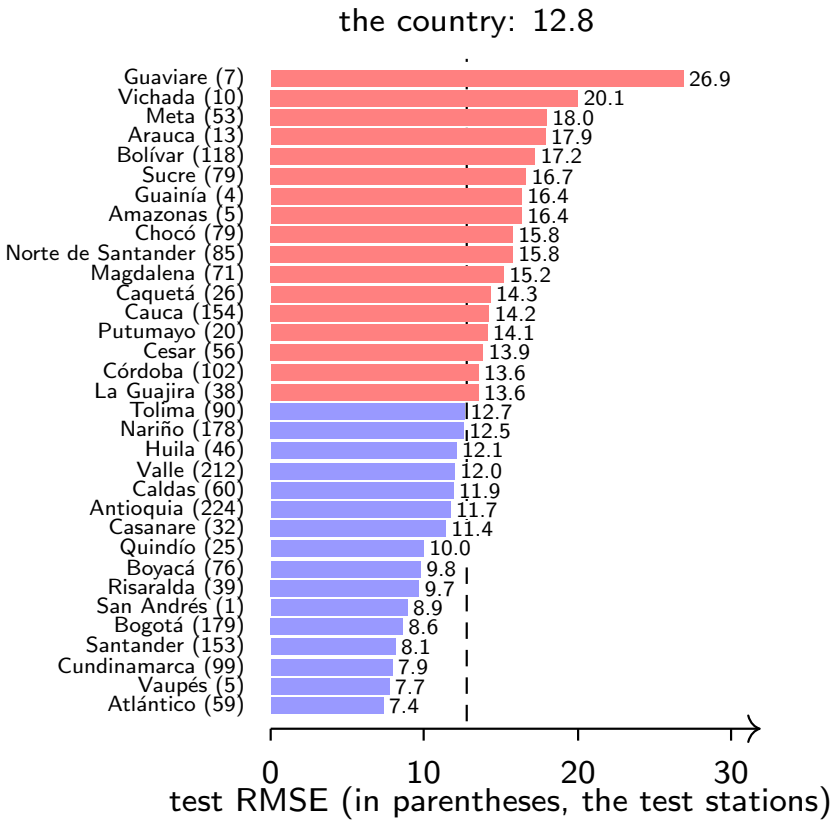
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The error by subgroup

The importances say what the prediction depends on; this says who it fails



- **By zone:** 9.3 in town centers and 14.9 in rural areas: the error lives in the countryside, where the census says less.
- **By vote level:** where Petro got less than 20, the model predicts 20.1 (actual: 12.5); where he got more than 80, 80.5 (actual: 90.1). The extremes, pulled toward the mean.
- **By department:** worst in Guaviare (26.9, with 7 stations), Vichada, and Meta; best in Atlántico (7.4) and Vaupés.
- **The rule:** without the error by subgroup the report is incomplete; if the error concentrates in one group, the average lies.

Fairness: when the subgroup is a person

- **The per-group metric.** False positives, false negatives, calibration, and precision *by group* (ethnicity, gender, stratum). A model with a high AUC can have twice the false negatives in one group. Our case allocates stations; a model allocating credit or subsidies allocates lives.
- **The label matters more than the algorithm.** Obermeyer et al. (2019): a health model used spending as a proxy for need; Black patients spend less at equal illness and were underestimated. Changing the label raised their share among the prioritized from 17.7 % to 46.5 %.
- **Dropping the sensitive variable is not enough.** Zip code, stratum, and name rebuild it: in our model, dropping the ethnicity block costs 0.3 points, because the department and the rest of the census rebuild it.
- **You cannot have it all.** Per-group calibration and equal error rates are incompatible when prevalence differs (Kleinberg, Mullainathan, and Raghavan, 2017). You must choose, and say which.

For your projects

If the prediction assigns something to people, the per-group metrics table goes next to the test table.

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Problem Set 2 is released today

From trees to the course's final model, with Cali's housing. Due: Tuesday, November 10, 11:59 p.m.

Same dataset and same rules as Problem Set 1: Cali's listings with their fixed split; the test price stays hidden and opens exactly once, for the course's final model.

Part	Session	Pts	What you do
1. Trees, forests, and boosting	5	20	CV-pruned tree; forest (*mtry*, OOB vs. validation); boosting with early stopping; table against PS1's best model
2. Opening the box	6	25	permutation vs. what the Lasso kept; PDP and ICE; global SHAP and two listings'; block ablation
3. The network against the forest	7	10	a small perceptron, early stopping, three seeds: does it beat the forest?
4. Final model and reading	5–7	15	the best model of both problem sets $\rightarrow$ test once; one page: what the model learned about the housing market and what it does not say
Course ranking		20	by the final model's test RMSE; first place 20, last 10
Reproducibility		10	`00_run.R` runs start to finish

Parts 1 and 2 are covered as of today; part 3, on October 27. **Debrief** of the PS1 ranking in session 7 and of PS2 on November 24.

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What the opened box says and what it does not

- **It says what the prediction depends on:** the department first and ethnicity next; the census blocks matter as a bundle: all used, almost none irreplaceable.
- **It says with what shape:** the Afro share pushes upward; education, nearly flat country-wide and with opposite slopes in Cali and Chocó.
- **It says which interactions it found:** above all ethnicity within each department: the slopes week 3 left unwritten.
- **It says where it fails:** in rural areas, at the extremes, and in departments where the census says little.
- **It does not say why,** nor what would happen if something changed, nor anything about the voter; and it changes with the model: with the map, another story with almost the same error.

Y, not β, with the box open

The Afro share's SHAP is not a β: it answers "how does the model use this variable to predict?". Moving to "what would happen if it changed?" demands an identification design: the Causal Inference course.

When to use what

	Use when	Watch out for
Gain or impurity	a free first look	cardinality; measured on the training set; never report it alone
Permutation	ranking variables in any model	extrapolates under correlation; unstable; shuffle whole blocks
Ablation	asking about one type of information	retrains; measures the irreplaceable, not the used
PDP and ICE	the shape of the effect	the average hides heterogeneity; extrapolates
ALE	the shape, with correlated variables	less intuitive; not local
SHAP	coherent local and global explanation	the fill-in; slow without trees; not causal
SHAP interactions	which interactions the model found	expensive: one matrix per station
Error by group	always	leaving it out

What to read: Molnar (2025), with code, for the permutation, PDP, ALE, and SHAP chapters; Lundberg and Lee (2017) for the unification; Lundberg et al. (2020) for TreeSHAP; Rudin (2019) for the objection.

Recap

1. **To interpret** is to ask the model how it predicts, always on data it never saw. Global (the model on average) and local (one station).
2. **Importance:** gain is free and biased (the noise ranked 24 of 39); permutation measures how much shuffling hurts (the department: +11.5); ablation, the irreplaceable (each census block, at most 0.4).
3. **Dependence:** the PDP averages, the ICE curves show the heterogeneity (education falls in Cali and rises in Chocó), and the ALE avoids the stations that do not exist.
4. **Shapley and SHAP:** the only exact split of a prediction; in Quibdó, 15.9 to ethnicity and 17.6 to the department. SHAP interactions say which crossings the machine found: ethnicity within each department.
5. **The explanation belongs to the model:** with the map, another story. And the error by subgroup always goes in: rural 14.9 against 9.3.

Getting ready for next class...

To get the most out of next session

This week's readings:

- ISL, sections 12.2 (principal components) and 12.4.1 (k-means); sections 10.1 to 10.3 and 10.6 to 10.7 (neural networks).
  Guide: figure 12.1 (the first two components) and 10.4 (the digits network) are the ones from class; 10.6 is the honest section: when to use deep learning.

Problem Set 2: released today; due Tuesday, November 10 at 11:59 p.m. Session 7 opens with the **Problem Set 1 ranking debrief**.

Next session — representing: today we saw that the census blocks stand in for one another. How much distinct information is in the 37 columns? Without a y, PCA compresses them into a few indices and k-means groups the stations into territory types. With a y, a neural network learns its own intermediate variables.

R application: the same stations and week 5's same model: gain, permutation, and ablation; the PDP, ICE, and ALE; SHAP with `predcontrib` and its interactions; the error by subgroup.

Essential references

- Molnar, C. (2025). *Interpretable Machine Learning* (3. ed.). https://christophm.github.io/interpretable-ml-book/
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- Breiman, L. (2001). Statistical Modeling: The Two Cultures. *Statistical Science*, 16(3), 199–231.
- Fisher, A., Rudin, C., & Dominici, F. (2019). All Models Are Wrong, but Many Are Useful. *JMLR*, 20(177), 1–81.
- Friedman, J. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232.
- Goldstein, A., Kapelner, A., Bleich, J., & Pitkin, E. (2015). Peeking Inside the Black Box: ICE. *JCGS*, 24(1), 44–65.
- Apley, D., & Zhu, J. (2020). Visualizing the Effects of Predictor Variables in Black Box Models. *JRSS-B*, 82(4), 1059–1086.
- Lundberg, S., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS*, 30.
- Lundberg, S., et al. (2020). From Local Explanations to Global Understanding with Explainable AI for Trees. *Nature Machine Intelligence*, 2, 56–67.
- Rudin, C. (2019). Stop Explaining Black Box Machine Learning Models for High Stakes Decisions. *Nature Machine Intelligence*, 1, 206–215.
- Gelvez, J. D., Cardiles, A., Martínez-González, E. F., & Muñoz, M. (2026). How Predictable Is an Election? Working paper. [The source of the case data.]

8 Appendix: going deeper

A1. Why the Shapley value is unique

A **game** is a function $v : 2^P \to \mathbb{R}$ with $v(\emptyset) = 0$ on the subsets of $P = \{1, \ldots, p\}$. A **splitting rule** assigns $\phi_j(v)$ to each player.

1. Unanimity games form a basis. For nonempty $T \subseteq P$, let $u_T(S) = \mathbb{I}\{T \subseteq S\}$. Any v is written uniquely as $v = \sum_T c_T\, u_T$: there are $2^p - 1$ games and $2^p - 1$ coefficients.

2. In a unanimity game the split is forced. Players outside T are null ($\phi_j = 0$); those in T are symmetric, and by efficiency they share $u_T(P) = 1$: $\phi_j(u_T) = 1/|T|$ for $j \in T$.

3. Additivity extends to every game: $\phi_j(v) = \sum_{T \ni j} c_T/|T|$. A single rule satisfies the four axioms, and expanding the c_T gives the marginal-contributions formula:

$$\phi_j = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[v(S \cup \{j\}) - v(S) \Big].$$

The Shapley decomposition of R^2 (Shorrocks, 2013): the game is $v(S) = R^2$ of the model with the blocks in S; with seven blocks, 2^7 models. It is the order-free version of part 1's ablation.

A2. Filling in the absent: marginal and conditional SHAP

A prediction's game needs $v(S)$ for every subset of variables. Two definitions:

$$v^{\text{marg}}(S) = \mathbb{E}_{X_{-S}}\left[\hat{f}(x_S, X_{-S})\right], \qquad v^{\text{cond}}(S) = \mathbb{E}\left[\hat{f}(x_S, X_{-S}) \,\middle|\, X_S = x_S\right].$$

- **Marginal:** sample X_{-S} from a background set without looking at x_S. True to the model: a variable $\hat{f}$ does not use receives exactly zero. The price: it evaluates $\hat{f}$ at impossible combinations (part 2's stations that do not exist).
- **Conditional:** sample X_{-S} given x_S. True to the data: no extrapolation. The price: a variable the model ignores receives credit if it is correlated with one it does use, and estimating the conditional is hard (Chen, Janizek, Lundberg, and Lee, 2020).
- TreeSHAP has both variants: the "interventional" one, which is the marginal with a background set, and the "path-dependent" one, which approximates the conditional with the nodes' frequencies; `predcontrib` in `xgboost` uses the second. Report which one you used.

A3. KernelSHAP: Shapley as a weighted regression

Lundberg and Lee (2017): the Shapley value is the solution of a weighted linear regression. With $z \in \{0,1\}^p$ indicating which variables are "present" and $v(z)$ the expected prediction with only those,

$$\min_{\phi_0,\phi} \sum_z \pi(z) \left[v(z) - \phi_0 - \sum_j \phi_j z_j \right]^2, \qquad \pi(z) = \frac{p-1}{\binom{p}{|z|}\,|z|\,(p-|z|)},$$

with $|z|$ the number of ones.

- The weight π is huge for nearly empty or nearly full coalitions and small in the middle: it is exactly the one that makes the least squares match Shapley's formula.
- In practice a few hundred or thousand z's (out of 2^p) are sampled and the least squares is solved: approximate, model-agnostic, and slow, because each z requires evaluating $\hat{f}$ over the background set.
- **TreeSHAP** (Lundberg et al., 2020) avoids the sampling: it walks each tree keeping count of how many observations pass through each branch, and gets the exact ϕ_j's in time polynomial in the depth.

A4. PDP, marginal plot, and ALE, in formulas

For a predictor x_j and the rest x_{-j}, three curves answering different questions:

$$\text{PDP:} \quad f_{\text{PD}}(z) = \mathbb{E}_{X_{-j}}\big[\hat{f}(z, X_{-j})\big] \quad \text{(average over the marginal of } X_{-j}\text{)}$$

$$\text{Marginal:} \quad f_{\text{M}}(z) = \mathbb{E}\big[\hat{f}(X) \,\big|\, X_j = z\big] \quad \text{(average over the conditional)}$$

$$\text{ALE:} \quad f_{\text{ALE}}(z) = \int_{x_{\min}}^{z} \mathbb{E}\Big[\frac{\partial \hat{f}(X)}{\partial X_j} \,\Big|\, X_j = u\Big]\, du \;-\; \text{const.}$$

- The PDP extrapolates: it averages over values of X_{-j} that never occur with that z. The marginal plot does not extrapolate but it *mixes*: if x_j and x_k move together, x_k's effect sneaks into x_j's curve.
- The ALE averages the **derivative** conditional on $X_j = u$ (no extrapolation) and then **accumulates**; since it differentiates first, x_k's effect disappears. In practice, with K bands of x_j with edges $z_0 < z_1 < \cdots < z_K$:

$$\hat{f}_{\text{ALE}}(z) = \sum_{k=1}^{k(z)} \frac{1}{n_k} \sum_{i:\, x_{ij} \in (z_{k-1}, z_k]} \Big[\hat{f}(z_k, x_{i,-j}) - \hat{f}(z_{k-1}, x_{i,-j})\Big],$$

centered at zero. With independent predictors, the three curves coincide.

A5. Conditional and block permutation, and Rashomon importance

- **Conditional permutation** (Strobl et al., 2008): permute x_j only *within* groups defined by the variables correlated with it. It measures x_j's importance *given* x_{-j}, like a partial coefficient; the unconditional one measures the marginal.
- **Block permutation:** permute a block's columns together (the five ethnicity ones; the department's 33). It answers "how much does the model lose without this type of information?" and avoids the split between twins.
- **Ablation** (*leave-one-covariate-out*): refit without x_j or without the block and compare errors. No extrapolation, but it costs one fit per block and measures something else: the loss when *another* model replaces x_j with its proxies.
- **Model class reliance** (Fisher, Rudin, and Dominici, 2019): the range of x_j's importances across all near-optimal models (the Rashomon set). If it includes zero, no data forces using x_j: "the model needs ethnicity" speaks of this model, not of the problem.
- **Shapley regressions** (Joseph, 2019): regress y on each variable's Shapley values to give them a standard error. They say whether the contribution aligns with the outcome, not whether it causes it.

A6. SHAP interaction values and Friedman's H

SHAP interaction (Lundberg et al., 2020): a pair's Shapley value, with the second difference instead of the first,

$$\phi_{jk} = \sum_{S \subseteq P \setminus \{j,k\}} \frac{|S|!\,(p - |S| - 2)!}{2\,(p-1)!}\,\nabla_{jk}(S),$$

$$\nabla_{jk}(S) = v(S \cup \{j,k\}) - v(S \cup \{j\}) - v(S \cup \{k\}) + v(S).$$

- $\nabla_{jk}(S) = 0$ if j's contribution does not depend on whether k is in: no crossing, no shared credit.
- The split is completed with the own effect, $\phi_j = \phi_{jj} + \sum_{k \neq j} \phi_{jk}$. TreeSHAP computes it exactly (`predinteraction = TRUE`): one $p \times p$ matrix per station, hence today's 500 stations.
- **Friedman and Popescu's H (2008)** is the global version built from PDPs:

$$H^2_{jk} = \frac{\sum_i \left[\hat{f}_{jk}(x_{ij}, x_{ik}) - \hat{f}_j(x_{ij}) - \hat{f}_k(x_{ik})\right]^2}{\sum_i \hat{f}_{jk}(x_{ij}, x_{ik})^2},$$

with the curves centered: zero is additivity; one, pure interaction.

A7. SHAP for gana_petro: the log-odds scale

- Week 5's classification boosting (census and department; test AUC 0.941) predicts on the **log-odds** scale: the base is 0.13 and the SHAP values add up there,

$$\log \frac{\hat{p}(x)}{1-\hat{p}(x)} \;=\; \phi_0 + \sum_{j=1}^{p} \phi_j(x), \qquad \hat{p}(x) = \frac{1}{1+e^{-(\phi_0+\sum_j \phi_j(x))}}.$$

- **What moves the probability that Petro wins the most:** department (1.29 in log-odds, on average), Afro share (0.44), and stratum 2 (0.34): the same hierarchy as in the vote.
- **Careful when translating:** the same ϕ_j moves the probability a lot near 0.5 and almost nothing near 0 or 1. It is week 4's logistic $\beta\, p(1-p)$ effect.

Appendix references and others cited

- Shapley, L. S. (1953). A Value for *n*-Person Games. En *Contributions to the Theory of Games II*, 307–317.
- Strobl, C., Boulesteix, A.-L., Zeileis, A., & Hothorn, T. (2007). Bias in Random Forest Variable Importance Measures. *BMC Bioinformatics*, 8, 25.
- Hooker, G., Mentch, L., & Zhou, S. (2021). Unrestricted Permutation Forces Extrapolation. *Statistics and Computing*, 31, 82.
- Obermeyer, Z., et al. (2019). Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations. *Science*, 366, 447–453.
- Robinson, W. S. (1950). Ecological Correlations and the Behavior of Individuals. *American Sociological Review*, 15(3), 351–357.
- Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017). Inherent Trade-Offs in the Fair Determination of Risk Scores. *ITCS 2017*.
- Strobl, C., Boulesteix, A.-L., Kneib, T., Augustin, T., & Zeileis, A. (2008). Conditional Variable Importance for Random Forests. *BMC Bioinformatics*, 9, 307. [A5]
- Friedman, J., & Popescu, B. (2008). Predictive Learning via Rule Ensembles. *Annals of Applied Statistics*, 2(3), 916–954. [A6]
- Shorrocks, A. F. (2013). Decomposition Procedures for Distributional Analysis: A Unified Framework Based on the Shapley Value. *Journal of Economic Inequality*, 11, 99–126. [A1]
- Chen, H., Janizek, J., Lundberg, S., & Lee, S.-I. (2020). True to the Model or True to the Data? *arXiv*:2006.16234. [A2]
- Joseph, A. (2019). Shapley Regressions: A Framework for Statistical Inference on Machine Learning Models. *Bank of England Staff Working Paper* 784. [A5]